

I. Problem Statement

Attendance Management System

II. Requirement

A. User Requirements

These describe what the users (teacher and students) need to accomplish.

Student Requirements

- ☐ Students must be able to record attendance by scanning a QR code.
- ☐ Students must be able to access the attendance page using their mobile devices.
- ☐ Students must be able to enter their name and auto fill student ID.
- ☐ Students must be able to submit their attendance quickly and once per session.
- ☐ Students must receive confirmation after successful attendance submission.
- ☐ Students must scan a currently valid QR code during the attendance session. (20 sec)

Teacher Requirements

- ☐ Teachers must be able to start and end an attendance session.
- ☐ Teachers must be able to display QR code for attendance collection, QR refresh 30sec.
- ☐ Teachers must be able to view attendance records.
- ☐ Teachers must be able to export attendance records to CSV/Excel.

B. System Requirements

1. Non-Functional Requirements

- ☐ Security : Only devices connected to school WiFi can access the system.
QR codes must expire automatically after 30 seconds
Expire QR won't be accepted for attendance submission.
- ☐ Performance: Attendance submission must complete within a few seconds.
QR code refresh every 30 sec
- ☐ Reliability : Attendance data must be stored accurately without duplication.
- ☐ Usability : The system must be simple and easy for students to use on devices.
QR code updates must occur automatically without requiring teacher
- ☐ Availability : The system must be available during class sessions.

2. Functional Requirements

- QR Generate
 - ☐ System generate QR code for attendance session
 - ☐ The system automatically refreshes QR every 30 seconds.
 - ☐ The system invalidates expired QR code.
- Network Verification :
 - ☐ check whether the user is connected to the school WiFi before allowing access.
 - ☐ Deny users that is not using school wifi
- QR Code Access :
 - ☐ students can scan a QR code to access the attendance page.
- Student Validation :
 - ☐ allow students to access the attendance form after successful validation.
 - ☐ allow students to enter their Student Name or ID.
- Student Input :
 - ☐ The system allows students to enter their name or Student ID.
 - ☐ The system shall provide Fill Student ID based on student name input.
- Device Verification :
 - ☐ retrieve the device ID from the user's device.
 - ☐ check the device has already submitted attendance for the current session.
 - ☐ prevent duplicate submissions from the same device.
- Attendance Recording :
 - ☐ store attendance records (Student ID, Student name, Major, Timestamp).
 - ☐ attendance is only recorded after successful validation of Student ID, QR code, WiFi connection, and Device ID.
 - ☐ store attendance status as "Present"/ "absence".
- Attendance Reporting :
 - ☐ allow teachers to view attendance records.
 - ☐ allow export of attendance data to CSV/Excel.
- Confirmation Message
 - ☐ display a success message after attendance has been recorded successfully.

III. Functional Decomposition

A. Abstraction

- scan a QR code to access the attendance system
- checks if the student is connected to school WiFi
- allows students to enter Student Name or ID
- Provide auto-fill for student ID if enter name.
- verifies Student ID with the database
- Capture Device ID during attendance submission

- prevents duplicate attendance from the same device
- records attendance details
- allows teachers to view attendance records
- allows export of attendance data (CSV/Excel)

B. Decomposition

1. QR & Network Module

- It creates and displays the QR code.
- QR expires every 30 sec.
- It finds the student's phone IP address when they scan the code.
- It checks if that IP address belongs to the school's Wi-Fi.
- It blocks students who are not using the school Wi-Fi and allows those who are.

2. Student Input Module

- It shows the attendance form.
- Accept student input (Name or Student ID)
- Provide real-time ID suggestions from the database (if enter name).
- Retrieve the student's name.

3. Validation Module

- check if students ID is valid
- Ensure that students did not double submitted attendance.
- Allow one submission per student per session.

4. Data Persistence Module

- Stores attendance record in CSV files.
- Organizes attendance data by date.
- Record Student ID , Name ,Major , Class , Time stamp.
- Save attendance data correctly when multiple students submit.

5. Notification Core Module

- Displays success message after attendance submit.
- Show error message for duplicate submission , invalid QR.

IV. Input/ Process/ Output Analysis

1. QR & Network Gateway

- (Input):
 - The professor started the attendance session.
 - Student Scan
 - Student device IP address
- (Process):
 - It creates the QR code
 - check if the QR Valid or expire.
 - checks if the students use the school Wi-Fi and then validates access.
 - QR refresh every 30 sec
- (Output) :
 - A QR code shown on the screen
 - Access allowed or blocked.

2. Student input

- (Input) : Student enter Name or ID
- (Process) :
 - Search Student Database
 - Display matching student suggestions
 - Auto-fill student ID
- (Output) : Student information (Name ,ID)

3. Data Management

- (Input) : student detail (Name and ID) and the current date and time
- (Process):
 - Check for duplicate submission
 - Validate before saving.
 - Store attendance Record.
- (Output) : Record successfully and updated attendance data

4. Notification Engine

- (Input) : Attendance results(success or error).
- (Process): Check submission and generate messages.
- (Output) :
 - Success or error message.
 - Error message (duplicate / invalid QR)

V. DFD

Database store

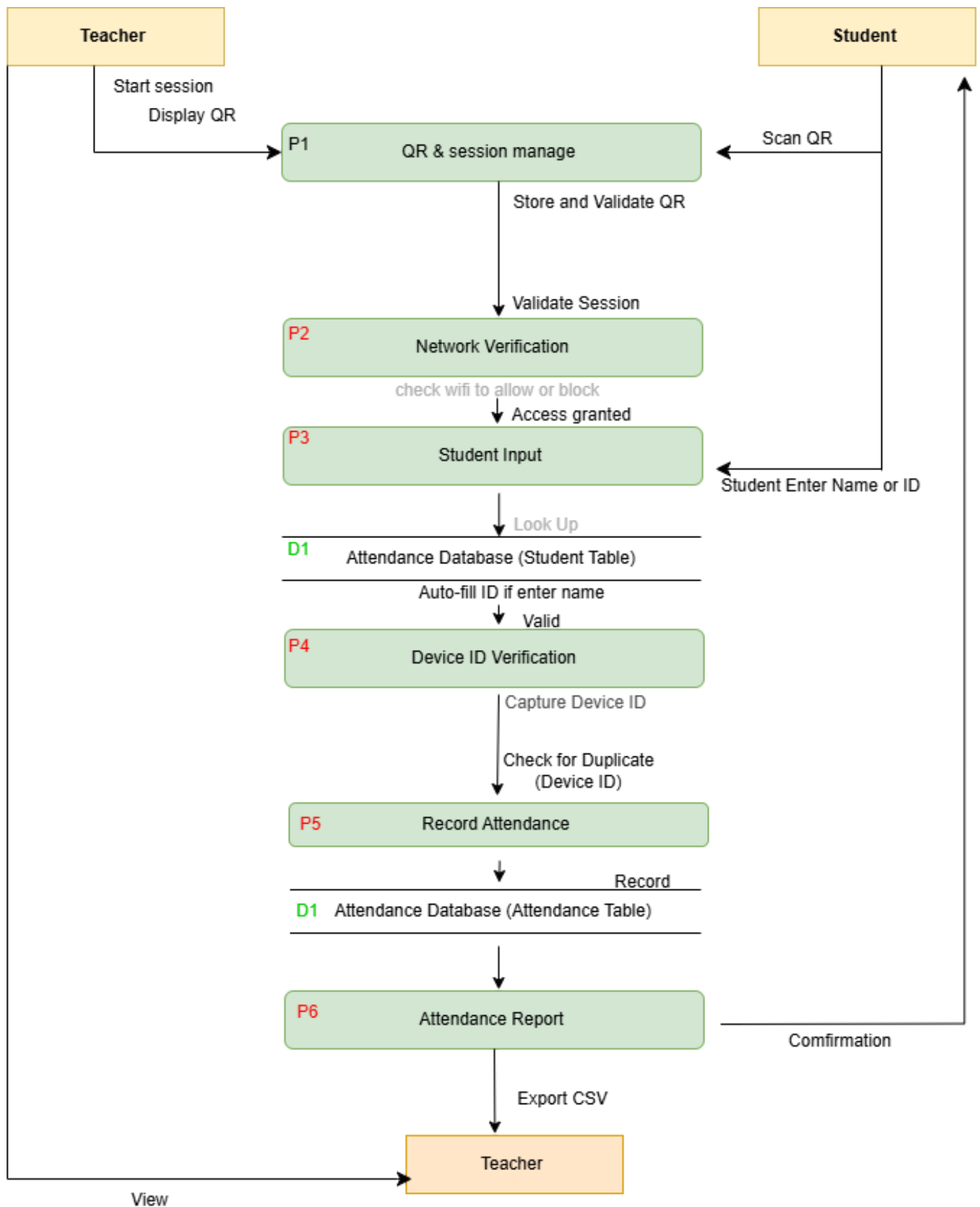
D1 Attendance Database: 2 table

1. **students** the roster
2. **attendance** the log of check-ins

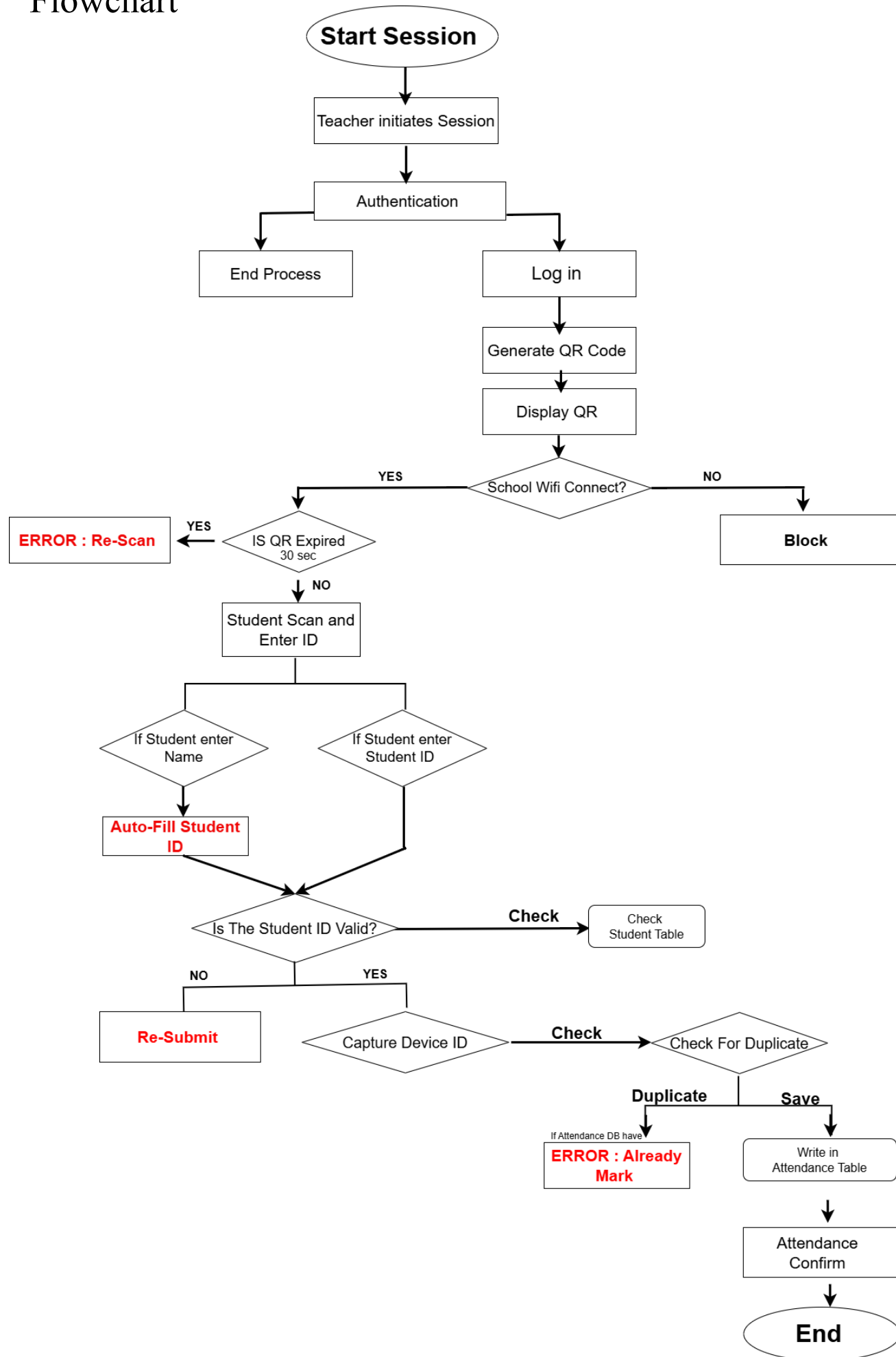
students table: **student_id** (primary key), **name**, **class_name**. This is your static roster — the autocomplete source.

attendance table: **id**, **student_id** (foreign key), **student_name** (denormalized, so the dashboard doesn't need a join to display names), **device_id** (a SHA-256 hash from the cookie), and **timestamp**. Major

Diagram Frow



VI. Flowchart



VII. Architecture system Design

